ASSIGNMENT 1 – CO1 | TASK 8

Read and Print Data Safely

Scenario: Namma Metro Smart-Card Fare System

Objective / explanation

Input extraction is checked before using the hour. If a non-number is entered, clear() resets the stream and ignore() discards the invalid line. Integer paise is formatted without floating-point rounding.

C++ source code

```
#include &lt;iomanip&gt;
#include &lt;iostream&gt;
#include &lt;limits&gt;
using namespace std;

int main() {
    int hour{};
    cout &lt;&lt; "Enter hour (0-23): ";
    if (!(cin &gt;&gt; hour)) {
        cin.clear();
        cin.ignore(numeric_limits&lt;streamsize&gt;::max(), '\n');
        cout &lt;&lt; "Invalid numeric input\n";
        return 0;
    }
    if (hour &lt; 0 || hour &gt; 23) {
        cout &lt;&lt; "Hour out of range\n";
        return 0;
    }
    long long paise = 1050;
    cout &lt;&lt; "Fare Rs " &lt;&lt; paise/100 &lt;&lt; '.' &lt;&lt; setw(2)
         &lt;&lt; setfill('0') &lt;&lt; paise%100 &lt;&lt; '\n';
}
```

Compilation

```
g++ -std=c++17 -Wall -Wextra -pedantic main.cpp -o program
```

Actual captured program output

```
Enter hour (0-23): Fare Rs 10.50
```

Output screenshot

Analysis / complexity

The shown output is for input 12. For letters, the program prints Invalid numeric input. setw applies to the next field; setfill controls padding.

Evidence note: Output shown is captured by executing this source in the generation environment. Recompile and run on your own system, and use your own screenshots if your instructor requires evidence from your computer.